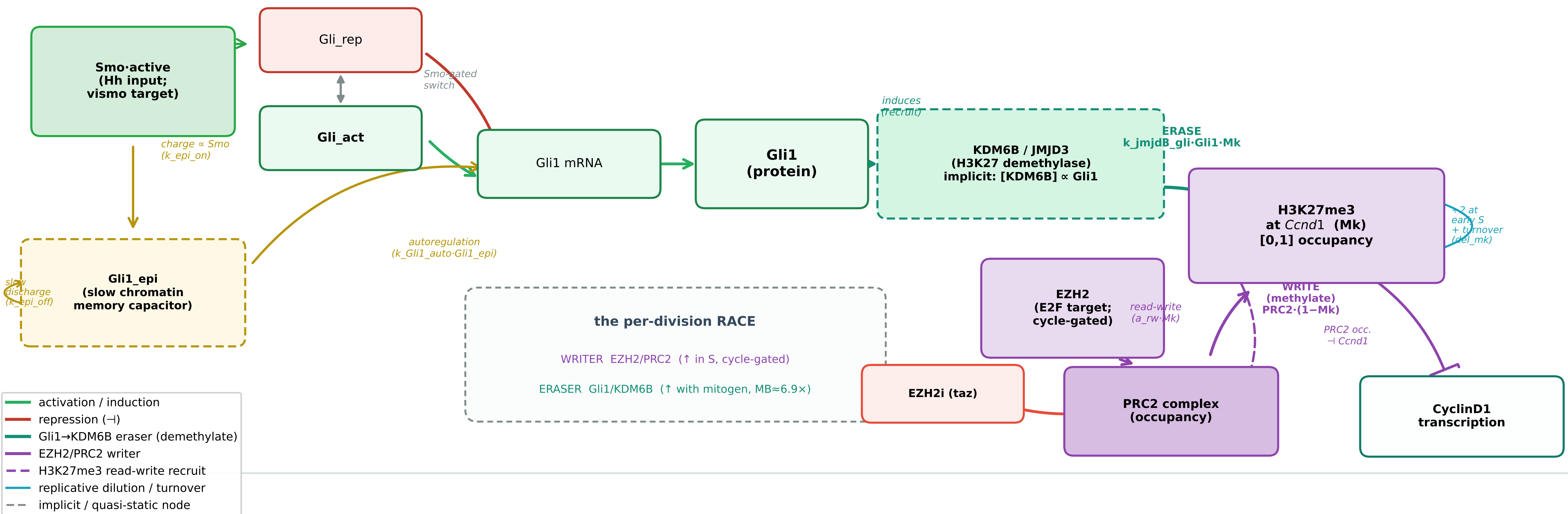
Gli1 → Kdm6b/Jmjd3 eraser of H3K27me3 at *Ccnd1* — detailed wiring + equations (v44.1)

mitogen-gated ERASER (Gli1 → KDM6B/JMJD3 demethylase) vs cycle-gated WRITER (EZH2/PRC2) — the per-division race that sets the mark



① Gli1 axis (mitogen input, with slow autoregulatory memory)

```
Gli_rep  ⇌  Gli_act        (Smo-gated activator/repressor switch)

d[Gli1_mRNA]/dt = Vmax·Gli_act²/(K_GA²+Gli_act²)·(1 − Gli_rep²/(K_GR²+Gli_rep²))·γ_Ptch
                  + k_Gli1_auto·Gli1_epi  −  k_mdeg·Gli1_mRNA

d[Gli1_epi]/dt  = k_epi_on·(1−Ptch)·Smo⁴/(K_auto⁴+Smo⁴)·(1−Gli1_epi) − k_epi_off·Gli1_epi

d[Gli1]/dt      = k_Gli1_tl·Gli1_mRNA − k_Gli1_deg·Gli1
```

② Eraser (KDM6B/JMJD3, implicit)

```
KDM6B is NOT a state variable.

Quasi-static:  [KDM6B] ∝ Gli1,

folded into k_jmjd3_gli.
```

erase flux =

k_jmjd3_gli · Gli1 · Mk

③ H3K27me3 (Mk) balance at the *Ccnd1* domain

```
d[Mk]/dt =  PRC2·(1 − Mk)   −   del_mk·Mk   −   k_jmjd3_gli·Gli1·Mk
           └ WRITE ────────┘  └ turnover ┘   └ ERASE ─────────┘
           (EZH2/PRC2, cycle)                (Gli1/KDM6B, mitogen)

discrete:  at Dna > 0.05  →  Mk ← ½·Mk        (replicative dilution, once per S)

PRC2 = EZH2·(1−EZH2i)·(a0 + a_rw·Mk)·(1 − g·Cd_mRNA^p/(K_tx^p+Cd_mRNA^p))
```

④ Baked parameters (v44.1) + net effect

```
k_jmjd3_gli 0.0478   del_mk 0.0015
a0 3.78e-4  a_rw 4.30e-3 (a_rw/a0 11.4)
g_prc2 0.0424  K_prc2 3.5e-3  n 3.39
f0_prc2 0.0505  K_tx 5.12  p_tx 3.27
Gli: Vmax 1.27  auto 0.012  on/off 0.02/8e-4
```

MB Gli1 ≈ 6.9× GNP → erase ≈ 6.9× stronger → ChIP Ccnd1 mark MB/GNP ≈ 0.55 (MB < GNP)

Net over divisions: Mk accumulates when WRITE > (turnover + ERASE + ½-dilution) each cycle; MB's high Gli1 tilts the race toward erasure despite higher EZH2.